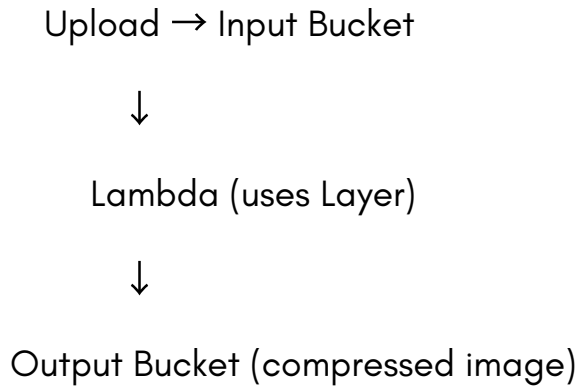


IMAGE COMPRESSION USING LAMBDA

ARCHITECTURE :



STEP-1 : Create S3 Buckets

- input-bucket-m4
- output-bucket-m4

STEP-2: Create IAM Role

- s3-acces --- putobject and getobject
- awsLambdaBasicExecutionRole

STEP-3: Create Lambda Function

- NAME: image-compressor
- RUNTIME: python.3.14

STEP-4: Add S3 Trigger

Go to Lambda → Triggers → Add

Select:

- **Bucket:** input-bucket-m4
- **Event:** PUT

STEP-5: On Local System

Create folder:

- `mkdir python`
- `cd python`

Install Python and Pillow Library

- `python --version`
- `python -m pip install pillow -t . --platform manylinux2014_x86_64 --only-binary=:all: --no-deps`
- zip the python folder as `pillow-layer.zip`

Final structure inside ZIP must be:

`pillow-layer.zip`

```
└── python/
    ├── PIL/
    └── Pillow-*.dist-info/
```

STEP-6: Creating Layer in Lambda

Go to **AWS Lambda**

1. Click **Layers (left sidebar)**
2. Click **Create layer**

Fill details:

- Name: `pillow-layer`
- Upload:
 - 👉 Select `pillow-layer.zip`
- Compatible runtime:
 - 👉 Select your Python version (e.g., Python 3.10)

Click **Create**

STEP-7 : Attach Layer to Lambda

1. Open your Lambda function
2. Scroll to **Layers section**

3. Click **Add a layer**
4. Choose
 - Custom layers
 - Select pillow-layer

✓ Done

STEP-8 : write code in lambda

```
import boto3

from PIL import Image

import io


s3 = boto3.client('s3')

OUTPUT_BUCKET = "output-bucket-m4"


def lambda_handler(event, context):

    bucket = event['Records'][0]['s3']['bucket']['name']

    key = event['Records'][0]['s3']['object']['key']

    response = s3.get_object(Bucket=bucket, Key=key)

    image_content = response['Body'].read()

    image = Image.open(io.BytesIO(image_content))

    # Compress image

    buffer = io.BytesIO()

    image.save(buffer, "JPEG", quality=50)

    buffer.seek(0)

    s3.put_object(

        Bucket=OUTPUT_BUCKET,

        Key="compressed-" + key,
```

```
Body=buffer,  
  
ContentType='image/jpeg'  
  
)  
  
return "Success"
```

- Deploy the code

FINAL OUTPUT :

- 👉 upload image to **input-bucket-m4**
- 👉 lambda triggers automatically , compresses image and uploads it into output-bucket-m4
- 👉 check the compressed image in **output-bucket-m4**